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Demonstration of an online digital twin for cost-effective operation of a water reuse pilot system

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Abstract

Modern water reuse systems face increasing operational complexity due to their energy intensity and tightening sustainability targets. This study presents the development and deployment of an online digital twin, operating in continuous interaction with a membrane-based water reuse pilot system. The digital twin is based on an empirical model that is updated hourly with the latest sensor data. A sparse interpretable machine learning model is used to forecast power consumption as a function of permeate, concentrate, and recirculation flowrates. An optimization routine determines energy-efficient operating setpoints under process and safety constraints at every hour. A receding-horizon implementation enables adaptive operation under changing conditions. Over a two-week (approximately 332-hour) deployment, the framework achieved an overall 38% reduction in specific energy cost relative to nominal steady-state operation, while maintaining stable actuator behavior. Temporal evolution of model parameters indicated convergence toward a compact and interpretable structure, demonstrating the robustness and stability of the digital twin during extended operation. These results establish the practical feasibility of digital twins based on empirical models for energy-aware, operational optimization in advanced water reuse systems.

Keywords

Water Reuse System; Energy efficiency; Desalination; Digital twin; Optimal production scheduling; Real-time closed-loop control; Adaptive L_1 -regularization; Pilot Study

Highlights

- A digital twin was developed for adaptive online optimization of a membrane-based water reuse pilot system.
- An empirical modeling and receding-horizon control framework enables hourly adaptive operation.
- A two-week (332-hour) deployment achieved an overall 38% reduction in specific energy cost.
- Stable coefficient evolution demonstrated robust and interpretable digital twin performance under dynamic conditions.

1 INTRODUCTION

Since the advent of water-quality sensor technologies in the 1980s, water production and reuse systems have become increasingly instrumented, generating large volumes of high-frequency operational data. Modern treatment facilities routinely collect thousands of sensor signals spanning flowrates, pressures, water quality indicators, and energy consumption. Historically, municipal wastewater treatment plants have leveraged these data to deploy advanced control strategies that address diurnal and seasonal variability in influent loading [1–3]. In contrast, water production and reuse systems have traditionally operated under near steady-state conditions, relying on buffer storage and relatively stable resource pricing to absorb operational variability. As a result, many facilities continue to operate at fixed setpoints despite growing external drivers for more flexible and adaptive operation.

Electricity price volatility has recently emerged as a significant operational constraint for advanced water reuse and desalination systems [1, 4–8]. Time-of-use tariffs introduce significant temporal variation in operating costs, yet most treatment trains remain insensitive to these signals, operating with static flowrate schedules that fail to exploit periods of lower electricity prices [9–11]. Previous studies have demonstrated the potential benefits of energy-aware operation in reverse osmosis (RO) systems, including improved integration with renewable energy sources and reduced electricity expenditure [12, 13]. In the present work, “energy-aware operation” refers specifically to operation that responds to time-varying electricity prices in order to reduce operating cost while maintaining required process performance. However, practical implementation of closed-loop control strategies in water reuse systems remains limited, and sustained online demonstrations under operational constraints are rare. Importantly, most water reuse systems are equipped with upstream and/or downstream buffer storage, enabling flexible operation without compromising water supply requirements. Within this flexibility, operational variables such as permeate, concentrate, and recirculation flowrates directly influence power demand in RO systems, providing practical control levers to reduce electricity costs under dynamic tariffs. In practice, however, this operational flexibility is constrained by equipment limitations, water quality requirements, and the need to maintain stable and reliable system performance.

Digital twins have emerged as a promising paradigm for enabling adaptive and data-driven operation of complex infrastructure systems. A digital twin is typically defined as a virtual representation of a physical asset that is continuously updated through data exchange with its physical counterpart, enabling monitoring, prediction, optimization, and control [1, 2, 14–17]. Continuous updating is particularly important in water reuse systems, where internal process dynamics such as membrane fouling, equipment aging, and nonlinear hydraulic behavior evolve over time. Unlike digital shadows, which support only one-way data flow from the physical system to the virtual model, digital twins entail bidirectional interaction that enables recommended control actions to be transferred back to the physical system [18–20]. Such bidirectional coupling is essential for predictive control and adaptive optimization in engineered systems exposed to continuously changing operating conditions [21, 22]. In safety-critical water treatment applications, however, such bidirectional interaction is typically embedded within a human-in-the-loop framework, where digital systems provide adaptive recommendations while trained operators retain supervisory authority over implementation.

Within the water sector, digital twins have been applied across a range of domains, including wastewater treatment, drinking water systems, pump scheduling, desalination, as well as water resources and distribution management [23–26]. Although these studies demonstrate the potential of digital twins for monitoring and decision support, many reported implementations remain offline, unidirectional, or simulation-based. As a result, there remains a gap between conceptual digital twin architectures and practical deployments [1, 5] capable of sustained autonomous operation under real-world operational constraints.

To address this gap, this work develops and deploys a bidirectional digital twin framework based on an empirical model-based optimization (EMBO) pipeline [27, 28]. The framework employs adaptive sparse regression to capture the relationship between controllable flowrates and power consumption and integrates this model within a receding-horizon optimization scheme that accounts for time-varying electricity tariffs. The approach is validated through a continuous two-week deployment on a pilot system operated at the Orange County Water District (OCWD) Advanced Water Purification Facility (AWPF). The pilot captures some key features of the advanced water reuse processes while remaining distinct from full-scale operation. By demonstrating reliable real-time operation, sustained cost reduction, and stable closed-loop behavior using conventional computational hardware, this study advances digital twin technology from conceptual

promise toward practical, scalable deployment for energy-aware water reuse operations.

2 MATERIALS AND METHODS

This section describes the three core components of the digital twin framework and the methods used to evaluate its performance: (i) the pilot system serving as the digital twin’s physical counterpart, representing a simplified physical model of the full-scale system; (ii) a data-driven empirical modeling and optimization pipeline forming the digital twin; (iii) the closed-loop feedback mechanism that couples the physical and virtual domains. Together, these components enable online adaptive optimization of system operation under hypothetical time-varying electricity tariffs. The electricity tariff used in this study reflects a time-of-use pricing structure designed to evaluate the potential benefits of energy-aware operation. It should be noted that currently OCWD AWPf operates under a relatively flat electricity rate, however dynamic pricing scenarios are considered here to assess future applicability and operational flexibility.

The digital twin operates according to a model predictive control (MPC) paradigm characterized by two principal horizons. An estimation horizon consisting of the most recent seven days of operational data is used for adaptive model calibration and parameter updating. A prediction or control horizon of 168 hours (one week) is used during optimization to evaluate tariff-weighted operating cost under anticipated conditions. At each hourly optimization step, the empirical model is retrained using the updated estimation horizon and a new 168-hour operating trajectory is computed. Only the first-hour control action from this optimized trajectory is implemented on the physical pilot, after which the optimization procedure is repeated using newly acquired measurements. This receding-horizon implementation enables continuous adaptation to evolving operating conditions and electricity tariffs.

Figure 1 illustrates the overall system architecture. Operational data collected from the pilot are stored in a cloud-based environment and supplied to the digital twin at hourly intervals, with each training window defined as the most recent seven days of operation. The pipeline couples an empirical model for power prediction with a cost-aware optimization module that computes flowrate setpoints over the 168-hour prediction horizon, subject to process and safety constraints.

2.1 Physical Pilot System and Measurements

Figure 2 illustrates the schematic of the physical pilot system used in this study, which represents key process components of OCWD’s advanced water reuse system though it does not rigorously replicate full-scale operation, which was not the study aim. For example, the full-scale facility is a three-stage RO configuration with RO concentrate discharge, whereas the pilot is a single-stage RO configuration with RO concentrate recirculation. Both approaches allow similar RO recoveries to be achieved (i.e., percent of purified water produced from the source water supply). In this sense, recirculation of concentrate in the pilot serves as an additional recovery across downstream RO stages in the full-scale facility, enabling comparable operating conditions to be studied within a compact configuration. The pilot was fed reclaimed water (i.e., filtered effluent from the adjacent wastewater treatment plant) provided by the full-scale facility, and served as the physical system on which to develop a digital twin.

In this framework, the pilot system (physical twin) provides real-time process and power data to the digital twin while receiving optimized control actions in return, enabling closed-loop operation for power/energy optimization. The main difference between a pilot and the physical twin is in the instrumentation layout that feeds the data necessary for the digital twin to carry out the optimization as envisioned. In this case, the presence of power meters provides data used for power/energy optimization by the digital twin. The pilot consists of three treatment units arranged in series: (i) ultrafiltration (UF); (ii) RO; (iii) ultraviolet (UV). The present work focuses on operational control within the RO unit, which is the dominant contributor to energy consumption and provides multiple controllable degrees of freedom.

Detailed pilot specifications, including UF and RO membrane models, pressure vessels, pump specifications, UV units, and the monitoring/control interface, are provided in the Appendix subsection A.1 (Figure A1 and Table A1). Briefly, the UF subsystem used polymeric hollow-fiber membrane elements for pretreatment, while the RO subsystem consisted of four 4 in. $\times$ 40 in. low-energy brackish-water RO membrane elements operated in a single-stage configuration with permeate, concentrate, and recirculation flow paths.

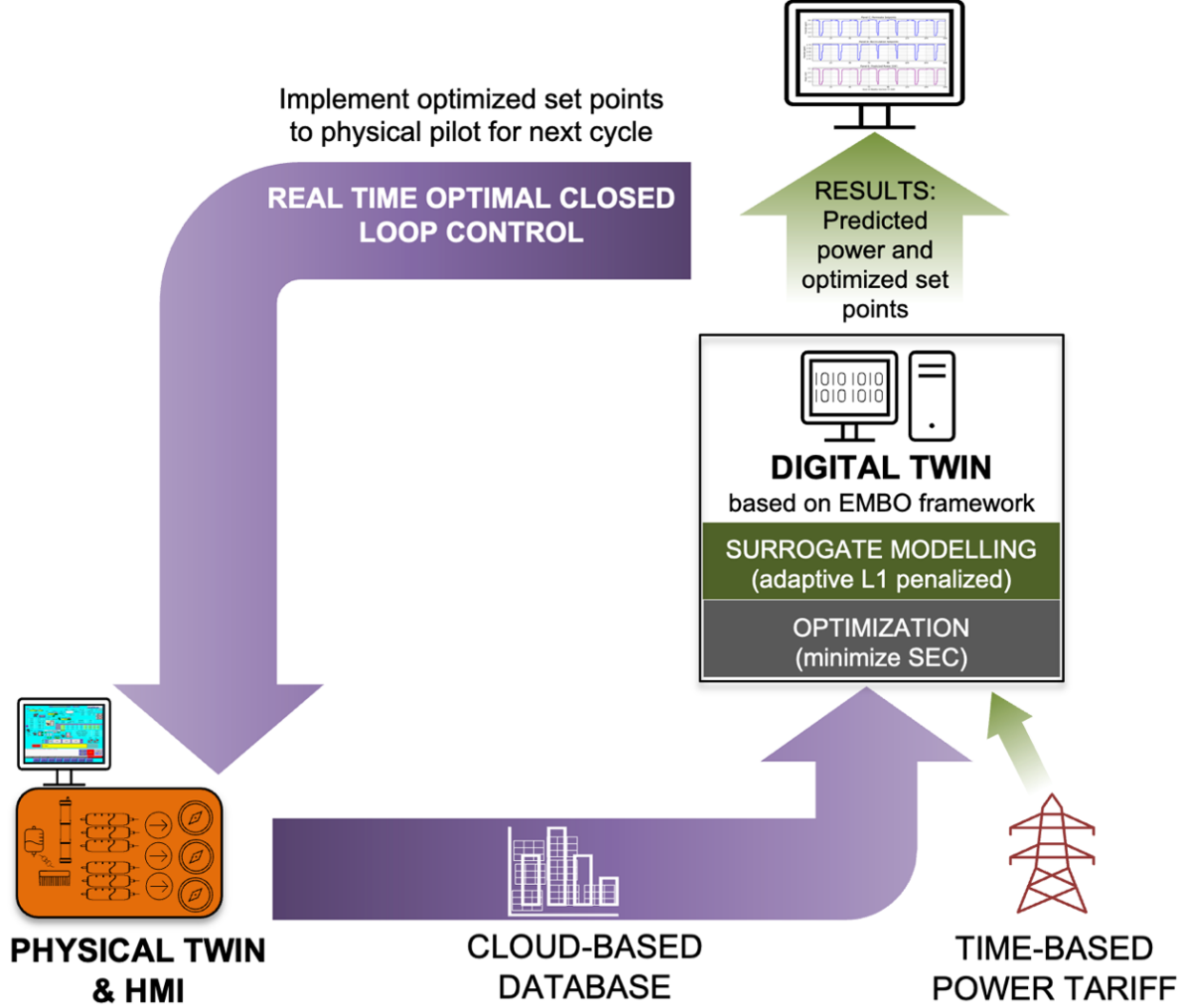


Figure 1: Digital twin framework for online energy-aware operation of a water reuse pilot. The pipeline integrates hourly empirical model calibration and cost-aware optimization over a 168-hour prediction horizon, with first-hour setpoints implemented on the physical system in a receding-horizon manner.

Within the RO subsystem, three flowrate variables are manipulated using automated valves: permeate (F_{perm}), concentrate (brine) (F_{conc}), and recirculation (F_{recyc}) flowrates (Figure 2). These flowrates can be adjusted through the pilot HMI (human-machine interface), remotely, or autonomously. F_{perm} was controlled through the RO pump/VFD and its associated feedback control loop, while the F_{conc} and F_{recyc} were controlled using automated pneumatic flow-control valves installed on the corresponding RO lines. These three flows determine all remaining internal flows, including membrane feed and reject streams, and therefore provide a minimal set of decision variables for energy-aware operation. The measurements used by the digital twin include online observations of F_{perm} , F_{conc} , and F_{recyc} , together with the aggregate electrical power demand of the pilot system, which serves as the measured response variable. Given the characteristics of a single-stage design, recirculation of concentrate in the pilot was necessary to achieve a level of recovery, commensurate with the full-scale facility.

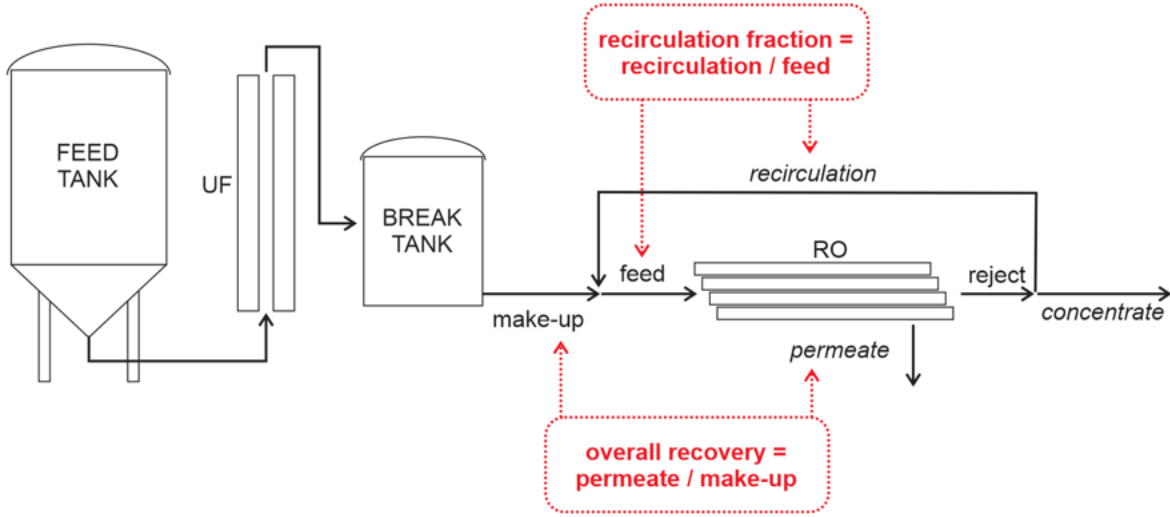


Figure 2: Physical pilot system comprising ultrafiltration (UF), reverse osmosis (RO), and ultraviolet and (UV) units. In this schematic flow configuration of the RO unit we highlight in red the permeate, concentrate, and recirculation flowrate decision variables.

2.2 Empirical Power Model and Exploratory Analysis

Figure 3 presents a model-free exploratory analysis of the measured data using a pairwise scatter-plot matrix. Each panel shows the relationship between two variables, while the reported Pearson correlation coefficients indicate the strength of their linear association. The analysis reveals a strong positive association between aggregate power demand and the total flowrate (Pearson correlation ≈ 0.83), as well as with permeate flowrate. Recirculation and concentrate flowrates exhibit moderate positive correlations. Based on inspection of these relationships, a nonlinear empirical model is developed to predict power draw from the flowrate signals. The observed structure of the data suggests that the flowrates provide informative predictors of system power consumption, and validation results indicate that this formulation yields reliable forecasting performance over seven-day training windows.

2.2.1 Regression Model Construction

Let $\mathbf{X} \in \mathbb{R}^{n \times 3}$ denote the input matrix comprising permeate, concentrate, and recirculation flowrates, and let $\mathbf{Y} \in \mathbb{R}^n$ denote the corresponding measured aggregate power demand. To capture nonlinearities and cross-coupling between decision variables, we construct a polynomial feature expansion up to cubic order ($d = 3$),

$$\Phi(\mathbf{X}) = [X_1, X_2, X_3, X_1^2, X_1X_2, X_1X_3, \dots], \quad (1)$$

yielding a library of 19 candidate basis functions. Because this representation can overfit when training data are limited (especially during early operation), model identification is performed using L_1 -regularized regression via the Least Absolute Shrinkage and Selection Operator (LASSO) [29–33], which promotes sparsity and interpretability. Specifically, regression coefficients β are estimated by:

$$\hat{\beta} = \arg \min_{\beta} \{ \|\mathbf{Y} - \Phi(\mathbf{X})\beta\|_2^2 + \lambda \|\beta\|_1 \}, \quad (2)$$

where $\lambda > 0$ controls the sparsity-accuracy trade-off. From a physical standpoint, the model is intended to capture the dependence of aggregate power demand on permeate, concentrate, and recirculation flowrates,

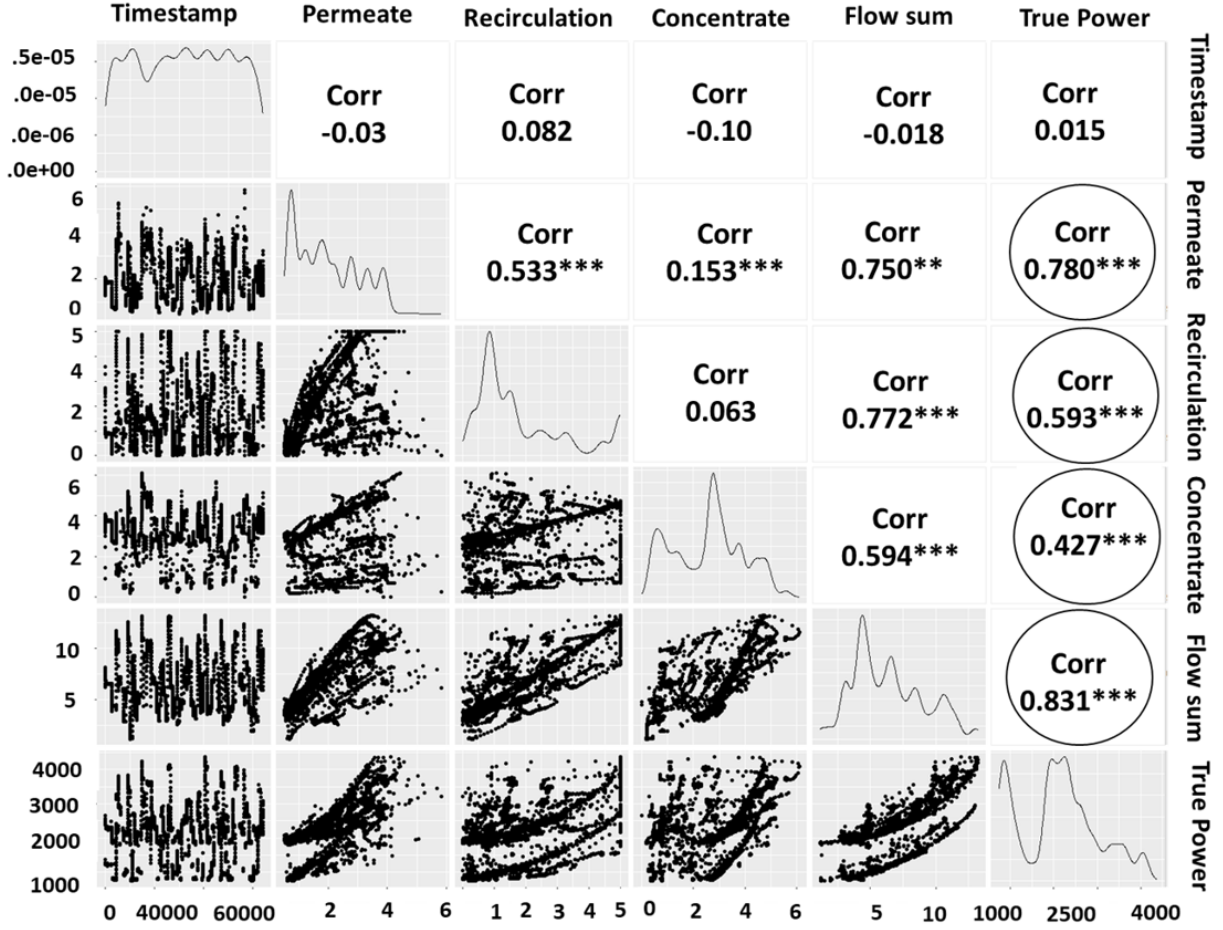


Figure 3: Pairwise scatter-plot matrix illustrating empirical relationships between aggregate power demand and flowrate variables. Pearson correlation coefficients demonstrate positive associations among power consumption and permeate, concentrate, and recirculation flowrates.

as changes in these variables affect the total flow handled by the RO system and, consequently, the pumping energy required. The polynomial expansion further allows this relationship to be nonlinear and to represent interactions among the manipulated flow variables.

2.2.2 Structured λ Selection with Degeneracy Safeguards

Model training uses 10-fold cross-validation [34–37] over a logarithmically spaced grid $\lambda \in [10^2, 10^{-4}]$. Let $\lambda_{\min}$ denote the regularization parameter that minimizes the cross-validated mean squared error (CV MSE), and let λ_{1se} denote the largest λ whose CV MSE lies within one standard error of the minimum. In early deployment, limited training data can yield unstable or overly sparse models (including degenerate solutions with all coefficients shrunk to zero). To ensure consistent model availability, EMBO employs a structured three-tier selection rule complemented by two degeneracy safeguards:

1. **Primary rule:** select $\lambda_{\min}$ if the number of non-zero coefficients at $\lambda_{\min}$ is ≤ 8 .
2. **Secondary rule:** otherwise scan from $\lambda_{\min}$ toward λ_{1se} and select the first λ yielding ≤ 7 non-zero coefficients.
3. **Fallback rule:** if no such λ exists, set $\lambda_{\text{opt}} = \lambda_{1se}$.

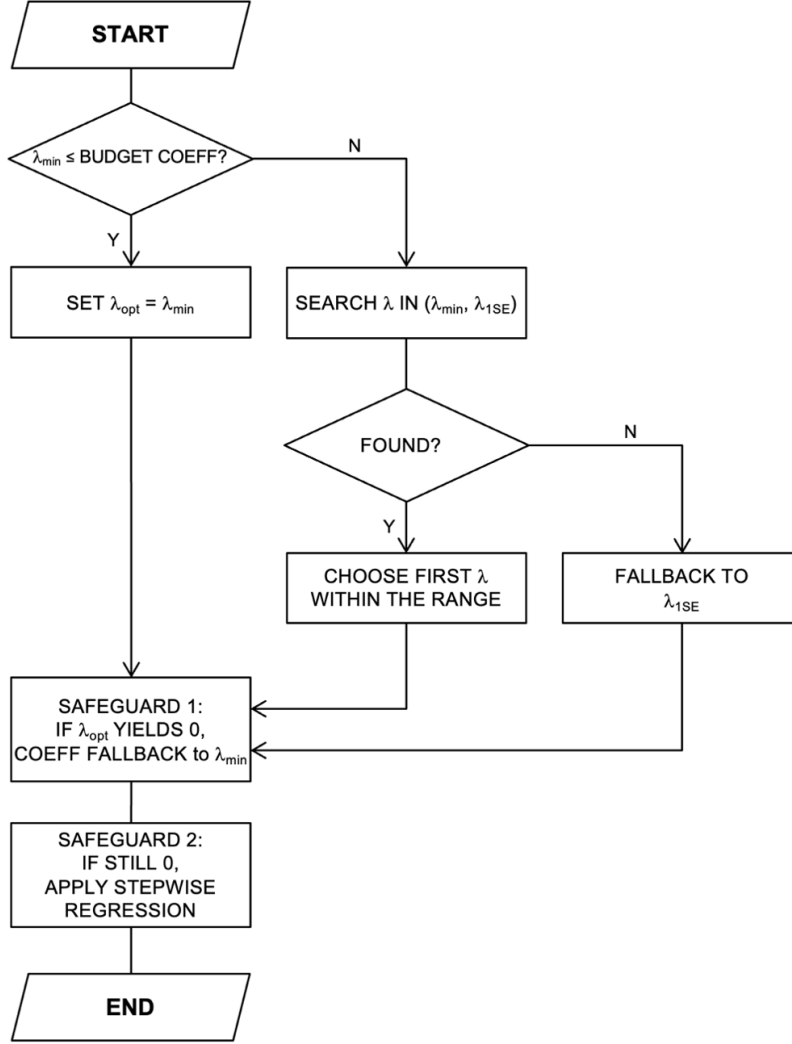


Figure 4: Schematic illustrating the adaptive L_1 -regularization strategy and structured fallback logic used for empirical model identification.

4. **Safeguard 1:** if $\lambda_{\text{opt}} = \lambda_{1\text{se}}$ yields a fully sparse model, revert to λ_{min} .
5. **Safeguard 2:** if λ_{min} also yields zero coefficients, a minimal linear model is constructed via forward stepwise selection on the original (non-polynomial) inputs to prevent control degeneracy.

The complete decision logic is summarized in Figure 4.

2.3 Empirical Feasibility Constraints for Safe Setpoint Generation

For safe closed-loop operation, it is necessary to define an operating region that is both experimentally realizable and consistent with equipment limitations. An initial experimental campaign generates candidate setpoints using Latin hypercube sampling within the operational bounds specified in the pilot system's manual. During inspection of the recorded data, several setpoint combinations, while satisfying prescribed safety limits, cannot be physically realized. In particular, the single pump in the RO unit is unable to consistently deliver certain requested membrane feed flowrates (equipment constraint).

In practical terms, these limitations arise from equipment constraints such as pump capacity and hydraulic coupling within the RO system, which prevent certain combinations of flowrates from being consistently

achieved. The feasible operating region therefore represents flow conditions that can be reliably implemented by the system under stable operation.

To account for this limitation, we identify the subset of experimentally feasible operating points and derive linear inequality constraints describing their convex envelope. These feasibility constraints are incorporated into the downstream optimization problem to prevent the digital twin from proposing operationally unattainable setpoints.

To quantify realizability, we segment the experimental time series into steady-state blocks in which the setpoint flowrates ($F_{\text{perm,set}}, F_{\text{conc,set}}, F_{\text{rcyc,set}}$) remain constant. Within each block, we compute median measured flowrates ($F_{\text{perm,med}}, F_{\text{conc,med}}, F_{\text{rcyc,med}}$) to reduce sensitivity to transients and measurement noise. To measure the closeness of the observed flow rates to the requested flow rate setpoint combination, the maximum absolute deviation (an ℓ_∞ metric) is defined as $d_{\text{max}} = \max \left\{ |F_{\text{perm,set}} - F_{\text{perm,med}}|, |F_{\text{conc,set}} - F_{\text{conc,med}}|, |F_{\text{rcyc,set}} - F_{\text{rcyc,med}}| \right\}$.

Smaller d_{max} indicates high setpoint fidelity, whereas larger values suggest infeasibility due to actuator limitations or coupling effects. Figure 5(a) visualizes all setpoints in 3D, colored by d_{max} . Based on the empirical distribution of deviations, the feasible region is defined as $\mathcal{F} = \{\mathbf{F}_{\text{set}} \in \mathbb{R}^3 \mid d_{\text{max}} < 0.2\}$, where $\mathbf{F}_{\text{set}} = (F_{\text{perm,set}}, F_{\text{conc,set}}, F_{\text{rcyc,set}})^\top$ and the 0.2 gallon per minute (gpm) threshold reflects a conservative tolerance under which tracking is consistently reliable.

To obtain a compact representation suitable for constrained optimization, the 3D convex hull of feasible setpoints is computed using standard computational geometry algorithms. Each hull facet defines a supporting half-space

$$a_{\text{perm}}F_{\text{perm}} + a_{\text{conc}}F_{\text{conc}} + a_{\text{rcyc}}F_{\text{rcyc}} \leq b, \quad (3)$$

and the collection of these inequalities defines a polyhedral approximation of the empirically realizable operating region. The complete list of hull-derived inequality coefficients is provided in the Appendix subsection A.2 (Table A2).

Figure 5(b) shows the 3D setpoints colored by whether they satisfy all hull-derived half-spaces. Points satisfying all convex-hull inequalities are marked in green, while violations appear in red. For additional physical interpretability, we project setpoints into aggregate flow coordinates $F_{\text{feed}} = F_{\text{perm}} + F_{\text{conc}} + F_{\text{rcyc}}$ and $F_{\text{reject}} = F_{\text{conc}} + F_{\text{rcyc}}$, and visualize the corresponding 2D feasibility envelope (Figure 5(c)). Finally, Figure 5(d) depicts the polyhedral convex hull itself, highlighting boundary points (in purple) that generate the hull facets and thus directly correspond to the linear constraints used during the optimization process.

2.4 Digital Twin for Online Optimization

At each hourly control cycle, the digital twin updates the empirical power model using the most recent estimation horizon and computes a 168-hour flowrate schedule that minimizes tariff-weighted specific energy cost while meeting production targets and operational constraints. Only the first-hour setpoints are implemented on the pilot via the programmable logic controller (PLC), after which new measurements are acquired and the procedure repeats, yielding receding-horizon operation.

2.4.1 Power Forecasting and Cost Evaluation

Given a candidate flowrate trajectory $\mathbf{X}_{\text{cand}} \in \mathbb{R}^{168 \times 3}$ over the 168-hour prediction horizon, predicted power demand is computed as $\hat{\mathbf{Y}} = \Phi(\mathbf{X}_{\text{cand}})\hat{\boldsymbol{\beta}}$. The sparse structure of $\hat{\boldsymbol{\beta}}$ enables interpretability by identifying dominant flowrate terms and interactions.

$$J(\mathbf{X}_{\text{cand}}) = \sum_{t=1}^{168} \hat{Y}_t c_t.$$

Figure 6 illustrates feature importance derived from the model after eight hours of operation, where only 10 of the 19 candidate terms remain active (approximately 47% sparsity). First-order flowrate terms dominate the learned mapping, indicating physically consistent sensitivity of power demand to operational setpoints.

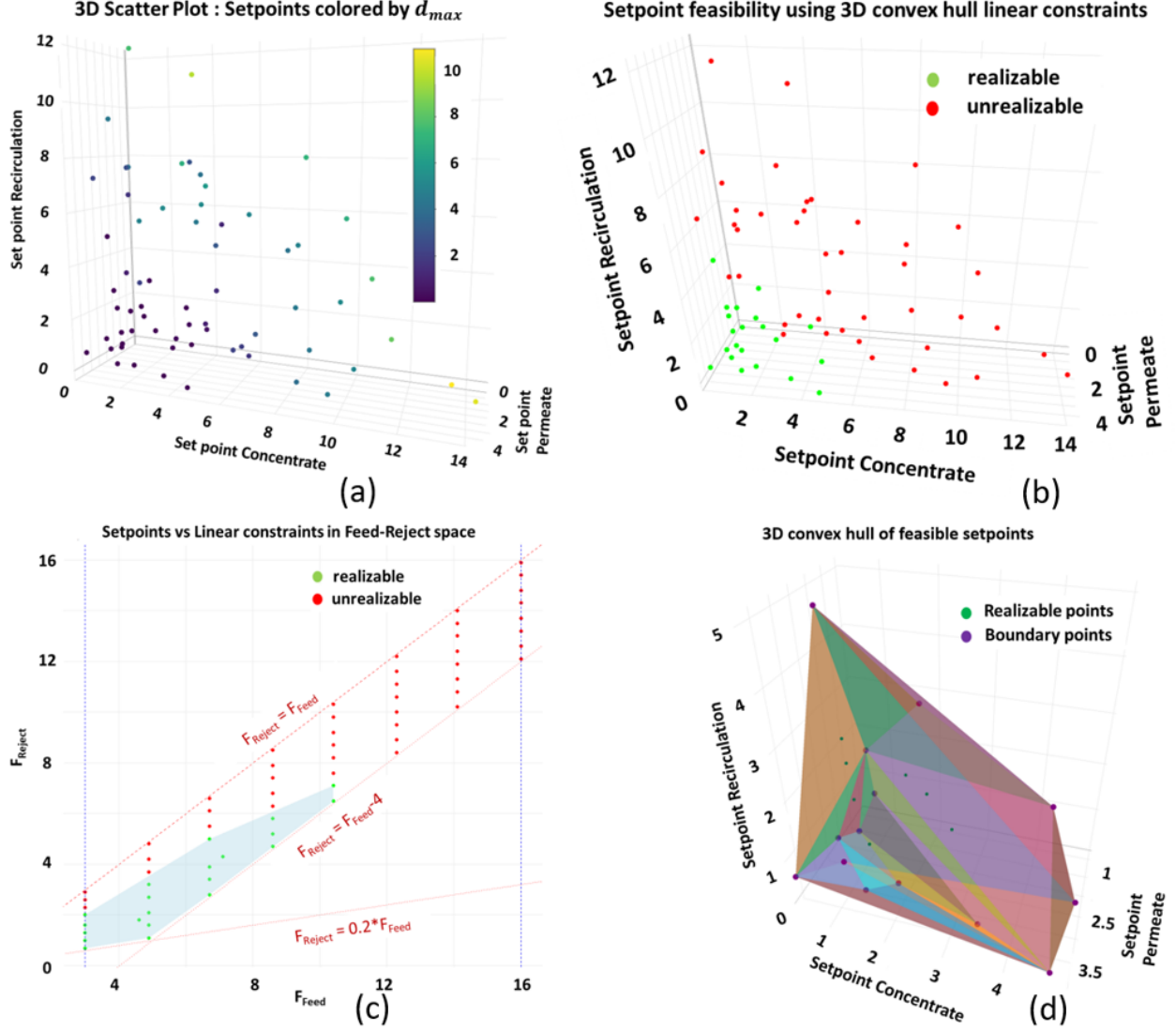


Figure 5: (a) 3D view of setpoint realizability colored by $d_{\max}$. (b) Setpoints colored by feasibility under hull-derived linear constraints. (c) Projection into feed–reject space. (d) Convex polytope and boundary points defining the hull facets.

2.4.2 Optimization Formulation

Tariff-weighted cost is evaluated using hourly electricity tariffs $\mathbf{c} \in \mathbb{R}^{168}$. The decision variable is the hourly flowrate trajectory $\mathbf{X}_{\text{cand}} = \{F_{\text{perm}}^{(t)}, F_{\text{conc}}^{(t)}, F_{\text{rcyc}}^{(t)}\}_{t=1}^{168}$, yielding a $24 \times 7 \times 3 = 504$ -dimensional optimization problem. The optimization is posed as

$$\min_{\mathbf{X}_{\text{cand}} \in \mathbb{R}^{168 \times 3}} \sum_{t=1}^{168} \left(\Phi(\mathbf{X}_{\text{cand}}^{(t)}) \hat{\beta} \right) c_t, \quad (4)$$

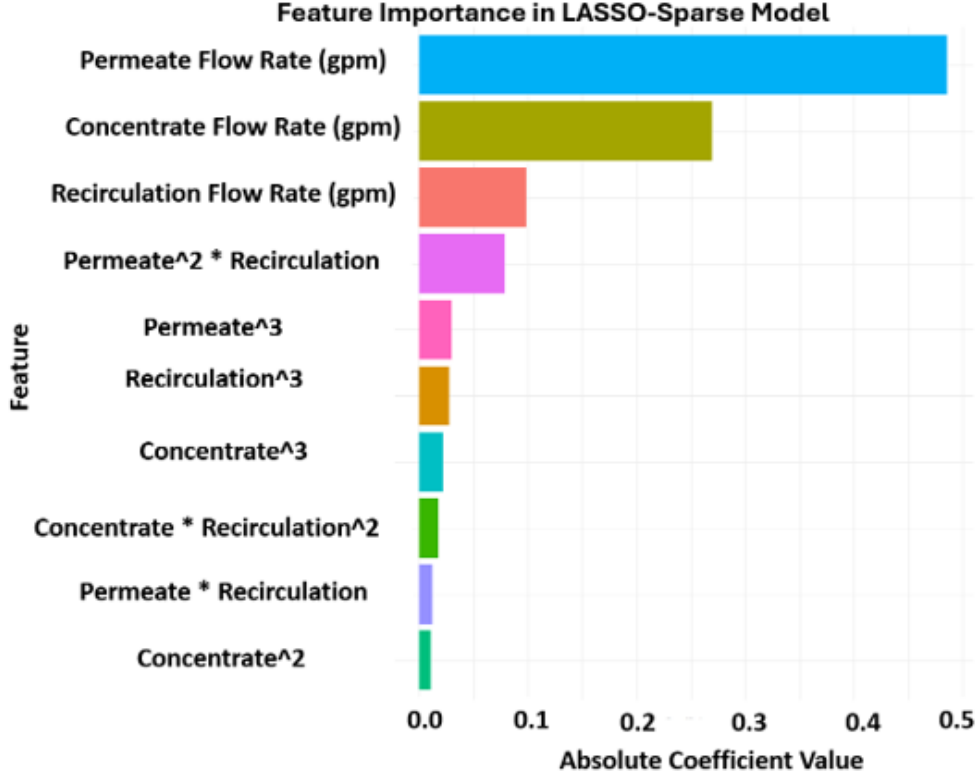


Figure 6: Feature importance derived from the sparse empirical model based on LASSO after 8 hours of operation.

subject to equipment and operational constraints, including:

$$0 \leq F_{\text{perm}}^{(t)} \leq 4, \quad (5)$$

$$3 \leq F_{\text{perm}}^{(t)} + F_{\text{conc}}^{(t)} + F_{\text{rcyc}}^{(t)} \leq 16, \quad (6)$$

$$F_{\text{perm}}^{(t)} \leq 4 \left(F_{\text{conc}}^{(t)} + F_{\text{rcyc}}^{(t)} \right), \quad (7)$$

$$(1 - R_{\text{target,h}})F_{\text{perm}}^{(t)} - R_{\text{target,h}}F_{\text{conc}}^{(t)} - R_{\text{target,h}}F_{\text{rcyc}}^{(t)} \leq 0, \quad \forall t, \quad (8)$$

$$\sum_{t=1}^{168} F_{\text{perm}}^{(t)} \Delta t = V_{\text{target}}, \quad (9)$$

$$(1 - R_{\text{target}}) \sum_{t=1}^{168} F_{\text{perm}}^{(t)} = R_{\text{target}} \sum_{t=1}^{168} F_{\text{conc}}^{(t)}. \quad (10)$$

Here $R_{\text{target,h}}$ denotes the hourly recovery target, R_{target} the weekly average recovery target, and V_{target} the required cumulative permeate volume.

The nonlinear constrained optimization is solved using the trust-region constrained method (**trust-constr**) implemented in SciPy [38–42]. The overall receding-horizon control architecture is illustrated in Figure 7 [43].

2.4.3 Post-Optimization Perturbation of Flowrate Schedules

In early operation or under weak tariff gradients, the optimizer may return nearly constant flowrate trajectories across the prediction horizon. If such trajectories are repeatedly implemented, the applied setpoints remain effectively constant over a complete estimation horizon, resulting in an uninformative dataset for

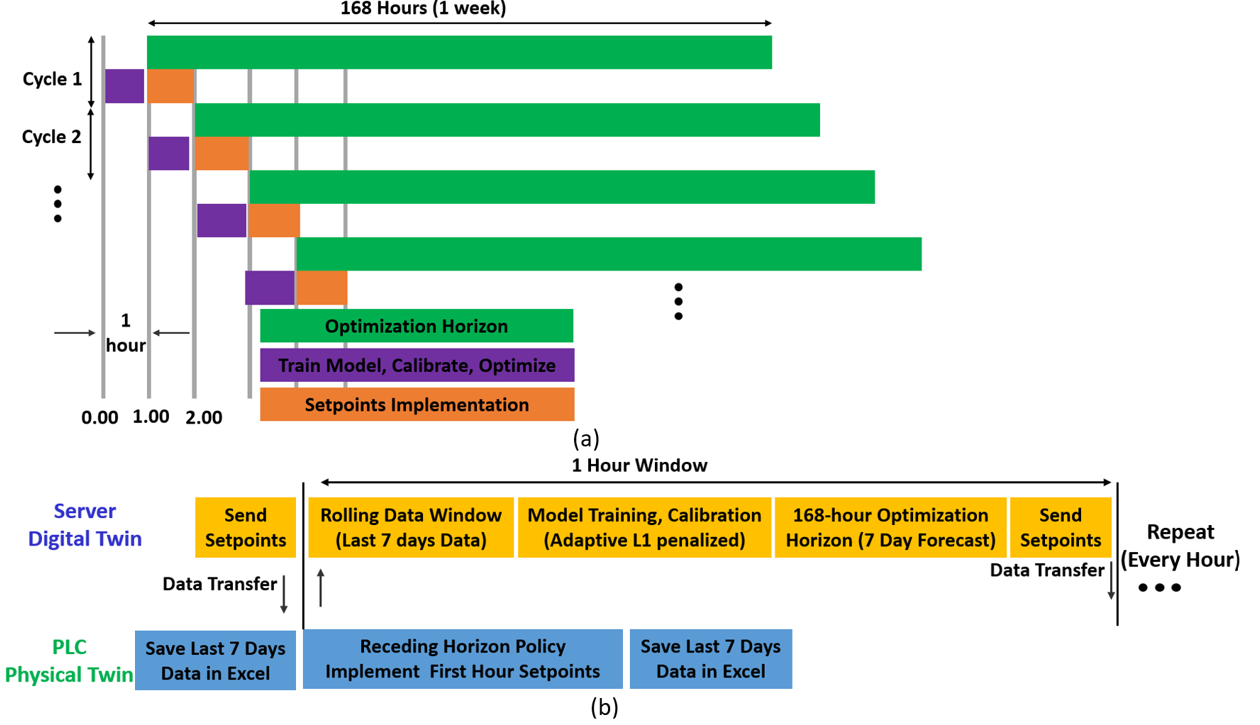


Figure 7: Digital twin control architecture: (a) overall closed-loop framework, (b) hourly receding-horizon control cycle.

subsequent model identification. To mitigate this issue while preserving operational feasibility, EMBO applies a bounded random perturbation to the optimized schedule. Let $\mathbf{Q}_{\text{opt}} \in \mathbb{R}^{168 \times 3}$ denote the optimized flowrate schedule and let $\mathbf{Q}_{\text{min}}, \mathbf{Q}_{\text{max}} \in \mathbb{R}^3$ denote the corresponding physical bounds. For each flow channel $j \in \{1, 2, 3\}$, the perturbation range is defined as $\Delta Q_j = \left(Q_{\text{max}}^{(j)} - Q_{\text{min}}^{(j)} \right) sf$, where s is a shrinkage fraction and f is a perturbation fraction. In the present implementation, the values $s = 0.05$ and $f = 0.98$ are used. A perturbation matrix $\mathbf{P} \in \mathbb{R}^{168 \times 3}$ is then sampled element-wise as $P_{tj} \sim \mathcal{U}(-\Delta Q_j, \Delta Q_j)$, and the perturbed schedule is obtained as $\mathbf{Q}_{\text{pert}} = \mathbf{Q}_{\text{opt}} + \mathbf{P}$.

After perturbation, setpoints are projected back onto the feasible region if necessary and rounded to enforce actuator-friendly updates prior to implementation. The 332-hour (approximately 2 weeks) dataset analyzed in this study represents a focused continuous validation period used to assess repeatability and sustained closed-loop performance of the digital twin. This validation campaign was preceded by an extended period of pilot operation, system integration, instrumentation checks, and iterative refinement, which was necessary to establish stable and reliable online deployment.

3 Results and Discussion

This section presents and discusses the results obtained from the two-week continuous experimental deployment of the proposed digital -physical twin framework, corresponding to approximately 332 uninterrupted hours of real time operation. The analysis focuses on the economic performance, operational behavior, and computational feasibility of the optimization under theoretical dynamic electricity tariffs.

The behavior of the structured λ selection rule during early deployment is illustrated in Figure 8. The figure shows that at eight hours of operation, cross-validation error bands tighten and conventional choices such as λ_{lse} yield stable sparse models (Figure 8 a–b). In contrast, after only one hour of operation, large uncertainty in cross-validation estimates can render λ_{lse} overly conservative, leading to very few active terms

(Figure 8 c–d).

Table 1 summarizes representative values of $\lambda_{\min}$, λ_{1se} , and the selected λ_{opt} , together with corresponding cross-validation mean square errors and sparsity levels. These results demonstrate that the structured selection rule prevents premature model collapse under data scarcity while preserving sparsity as training data accumulate.

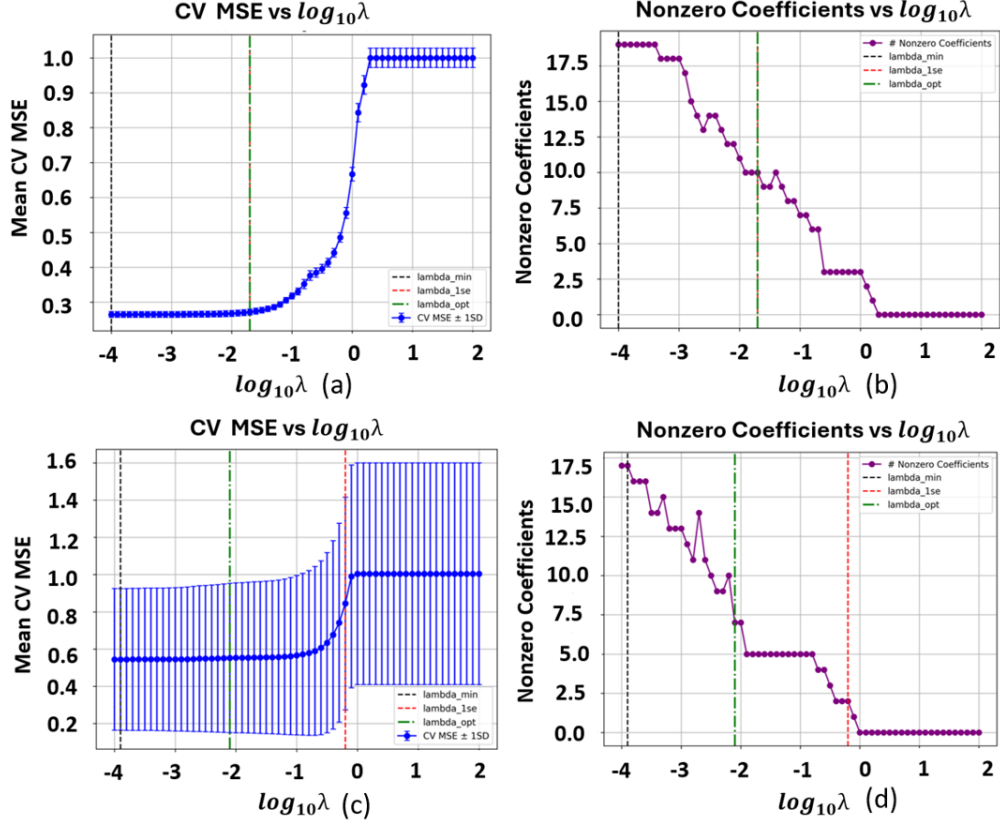


Figure 8: Cross-validation performance and sparsity behavior of the empirical model under different regularization levels: (a,b) results after 8 hours of operation; (c,d) results after 1 hour of operation.

Table 1: Representative cross-validation outcomes illustrating structured λ selection under limited (1 hour) and expanded (8 hours) training data.

Case	$\lambda_{\min}$	λ_{1se}	λ_{opt}	MSE $\lambda_{\min}$	MSE λ_{opt}	Nonzeros
8 hours	0.00009	0.0199	0.0199	0.2651	0.2722	10
1 hour	0.000125893	0.63095	0.00794	0.5445	0.8447	7

Figure 9 illustrates a representative optimization outcome selected at the 50th hour of deployment (approximately two days after initialization). This snapshot corresponds to a single 168 hour optimization horizon generated using the most recent model. As shown in Figure 9(a), the hourly electricity cost under optimized control is substantially reduced relative to nominal steady state operation, particularly during periods of elevated tariff. Importantly, the optimized flowrate schedules satisfy the same cumulative permeate production requirement over the 7-day horizon as the nominal operating policy, ensuring a fair comparison. Quantitatively, the optimized operating policy achieves a cost reduction of 41% relative to nominal operation for this representative horizon, as summarized in Table 2. The reduction reflects a decrease in specific energy cost driven by strategic load shifting while producing the same amount of water during the forecast horizon.

In practical terms, the optimized strategy reduces flowrates during periods of high electricity price and increases them during lower-price periods, thereby shifting production to lower-cost intervals while maintaining the same cumulative permeate output over the horizon.

Table 2: Cost comparison between nominal steady-state operation and optimized control for the representative 168-hour horizon shown in Figure 9.

Default Cost (\$)	Optimized Cost (\$)	Cost Savings (\$)	Cost Reduction (%)
198.15	116.14	82.01	41.39

Additional optimization examples corresponding to the 53rd and 300th hours of operation are provided in Appendix subsection A.3 (Figures A2 and A3), illustrating the repeatability and robustness of the developed framework across the full deployment period.

Figure 9(b) demonstrates how the predicted power demand adapts to the time varying tariff structure. The model driven controller systematically suppresses power consumption during high tariff periods while reallocating operational load to lower tariff intervals, confirming the intended economic responsiveness of the digital twin. This behavior is further reflected in Figures 9(c)–(e), which show the optimized 168-hour flowrate trajectories for concentrate, permeate, and recirculation streams. In contrast to the constant steady state setpoints used under nominal operation (shown in gray), the optimized flowrates dynamically decrease during high tariff intervals and increase during low tariff periods, consistent with cost aware operation across all manipulated variables. Throughout the experimental period, permeate water quality indicators, including electrical conductivity (EC) and total dissolved solids (TDS), remained within acceptable operating ranges, with EC consistently below approximately 250 $\mu\text{S}/\text{cm}$, well below the California State Water Resources Control Board guideline of 900 $\mu\text{S}/\text{cm}$. This indicates that the optimized flow variations did not adversely impact the purified water quality over the duration of the study.

Figure 9(f) illustrates the convergence behavior of the optimization routine for the same horizon. The objective function decreases monotonically and converges within a small number of iterations, typically requiring 5–10 minutes of computation on conventional hardware. This confirms that the full calibration optimization deployment cycle comfortably fits within the one hour control interval, enabling reliable real time operation.

Beyond this representative example, the cumulative performance over the full experimental campaign is summarized in Table 3. Across the entire 332-hour deployment, the proposed framework achieves an average cost reduction of 38% relative to nominal steady-state operation. The consistency of these savings over an extended period demonstrates the robustness and repeatability of the digital physical twin under sustained real time conditions.

Table 3: Aggregate cost performance over the full 332-hour deployment.

Default Cost (K \$)	Optimized Cost (K \$)	Cost Reduction (%)
67	41	38.3

Figure 10 illustrates the hourly flowrate setpoints implemented on the physical pilot throughout the full experimental deployment. At each control cycle, the digital twin recalibrates the empirical model using the most recent data and computes an updated 168-hour optimization horizon, of which only the first-hour setpoints are enacted on the pilot. The figure displays the applied optimized setpoints for concentrate (solid blue), permeate (solid green), and recirculation (solid red), together with the corresponding nominal steady-state flowrates (dashed lines). The results show a clear and consistent relationship between the applied setpoints and the time-varying electricity tariff. Across all three manipulated variables, the optimized setpoints are systematically reduced during high-tariff periods and increased during low-tariff periods, whereas the steady-state operation remains constant and unresponsive to tariff fluctuations.

The magnitude of cost savings is dependent on the assumed electricity tariff structure. In this study, a time-of-use electricity pricing scenario was used to evaluate the potential benefits of energy-aware operation; under other pricing structures, the economic impact may differ. For example, the OCWD AWPf where the

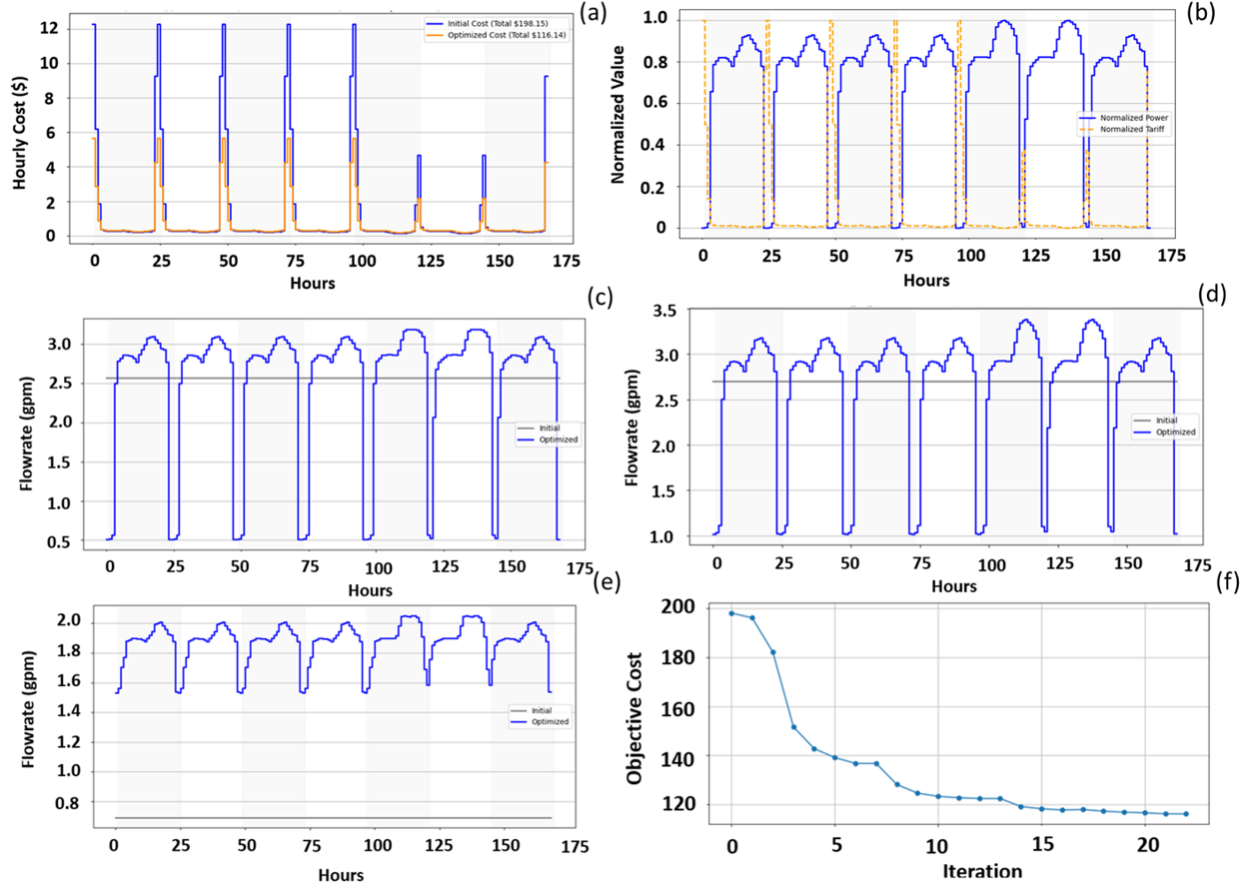


Figure 9: Representative optimization results at the 50th hour of deployment. (a) Hourly electricity cost under nominal steady-state operation (blue) and optimized control (orange), showing a cost reduction of 41.39%. (b) Predicted power demand (blue) overlaid with the normalized electricity tariff (orange), illustrating tariff-aware load shifting. (c) Concentrate flowrate profile under optimized control (blue) compared with steady-state operation (gray). (d) Permeate flowrate profile under optimized control (blue) compared with steady-state operation (gray). (e) Recirculation flowrate profile under optimized control (blue) compared with steady-state operation (gray). (f) Optimization convergence showing objective cost versus iteration number, demonstrating rapid convergence within the real-time control interval.

pilot was sited currently operates under a relatively flat electricity rate; therefore, at full-scale no energy savings could be achieved by dialing water production up or down at different times of the day or week. Dynamic pricing scenarios were considered here to assess potential future applicability should tariff structures change for a given utility, since numerous other utilities are already subject to time-of-use tariffs (e.g., the Water Replenishment District facilities in the Los Angeles Area).

To evaluate the interpretability and temporal stability of the digital twin, the evolution of regression coefficients over the full 332-hour experimental deployment was tracked. At each hourly control cycle, the trained LASSO model was archived, enabling post hoc analysis of coefficient trajectories and structural sparsity over time. This analysis provides insight into the convergence behavior of the model as training data accumulate under real-time operation.

Due to model training being performed on standardized data, the estimated regression coefficients $\hat{\beta}_j$ are expressed in normalized units. To recover physically interpretable coefficients β_j^{phys} (e.g., in units of W/gpm or W/gpm²), the coefficients are rescaled by reversing the affine transformations applied during standardization. Specifically, the physical coefficients are obtained as $\beta_j^{\text{phys}} = (\sigma_y / \sigma_{x_j}) \hat{\beta}_j$, where σ_y denotes

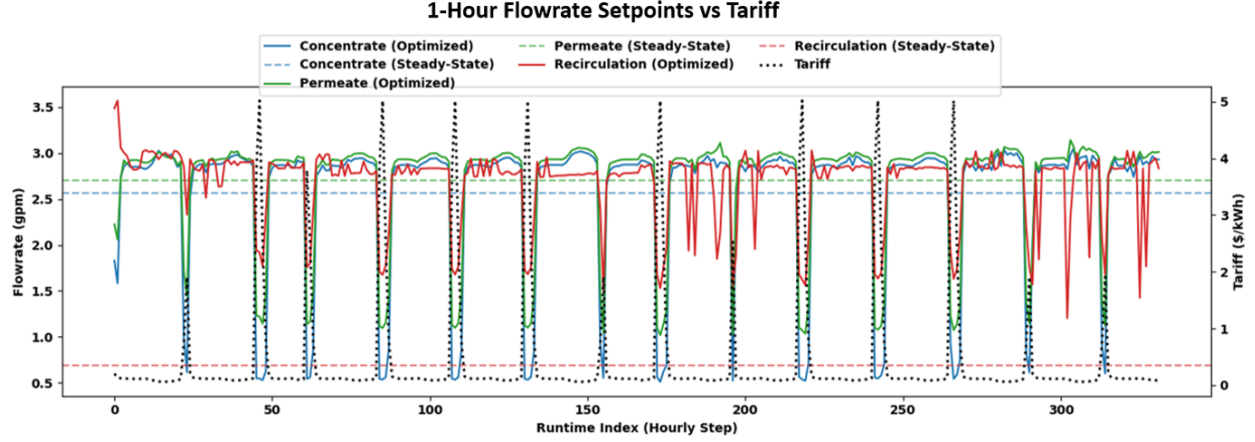


Figure 10: Hourly flowrate setpoints applied to the physical pilot over the full experimental deployment. Solid lines denote optimized setpoints implemented at each hour for concentrate (blue), permeate (green), and recirculation (red), while dashed lines indicate the corresponding steady-state operating values. The normalized electricity tariff is shown for reference. The optimized setpoints consistently decrease during high-tariff periods and increase during low-tariff periods, demonstrating stable and reliable tariff-aware control under continuous real-time digital twin operation.

the standard deviation of the measured power signal and σ_{x_j} denotes the standard deviation associated with the corresponding feature after polynomial expansion. This transformation maps the coefficients from normalized units back to their physical units.

Figure 11(a) summarizes the temporal evolution of all regression coefficients using a normalized heatmap representation. Each coefficient time series is normalized by its maximum absolute magnitude over the 332 hour window, $\hat{\beta}_j^{\text{phys}}(h) = \beta_j^{\text{phys}}(h) / \max_h |\beta_j^{\text{phys}}(h)|$, thereby rescaling all features to the interval $[-1, 1]$ for visual comparability. This representation highlights both coefficient activation patterns and relative temporal variability across features.

Figures 11 (b)–(d) further visualize the physical coefficients associated with the first-order flowrate terms: concentrate, permeate, and recirculation as explicit time series. Coefficients that were pruned by the LASSO regularization at a given hour are assigned a value of zero and shown in red, while active coefficients are shown in blue. This visualization directly illustrates the evolving structure under regular retraining.

Several consistent trends emerge from this analysis. First order flowrate coefficients remain persistently active throughout the deployment and exhibit relatively stable magnitudes after couple of hours, indicating convergence toward a reliable and physically meaningful model structure. In contrast, higher-order and interaction terms are activated intermittently, reflecting transient structural exploration during early learning stages followed by progressive sparsity enforcement as data volume increases. Certain higher order features appear only during specific temporal windows, suggesting localized relevance under transient operating conditions or tariff regimes. In later stages of the experiment, the number of active coefficients decreases and coefficient variability diminishes, indicating stabilization of model complexity and improved robustness of the learned empirical model.

To investigate convergence and structural stability in the optimized flowrate schedules produced by the proposed framework, the temporal consistency of the predicted 168-hour setpoint trajectories was analyzed using a rolling-aligned visualization. At each hourly optimization step t , the digital twin generates an optimized flowrate trajectory over the subsequent 168-hour prediction horizon for each manipulated variable (concentrate, permeate, and recirculation). For a given flowrate channel, this trajectory is denoted by $\mathbf{x}_t = [x_t^{(1)}, x_t^{(2)}, \dots, x_t^{(168)}] \in \mathbb{R}^{168}$, where $\mathbf{x}_t$ represents the predicted hourly setpoints starting at time t and extending over the next seven days.

Because the optimization is performed every hour, the first element of $\mathbf{x}_t$ corresponds to different hours within the weekly calendar across successive optimization steps. To enable meaningful comparison, we align

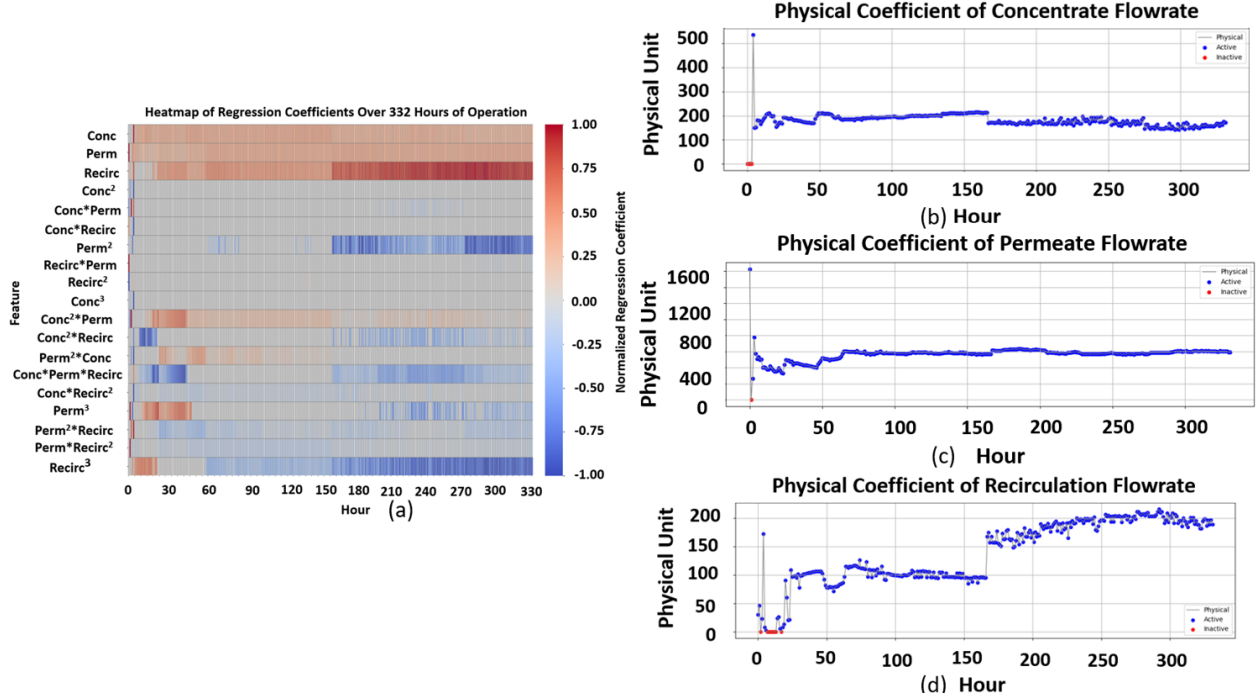


Figure 11: Temporal evolution of regression coefficients over 332 hours of real-time operation. (a) Heatmap of normalized physical regression coefficients, with each feature scaled to the interval $[-1, 1]$ based on its maximum absolute value across time. (b) Physical coefficient of the concentrate flowrate as a function of hour. (c) Physical coefficient of the permeate flowrate as a function of hour. (d) Physical coefficient of the recirculation flowrate as a function of hour. Active coefficients are shown in blue, while coefficients pruned by LASSO regularization are shown in red, highlighting structural sparsity and convergence of the model over time.

all predicted trajectories so that the first data point corresponds to the same hour-of-week in the weekly calendar, which in this study is defined as Monday at 12:00 AM.

Let $\mathbf{x}_t \in \mathbb{R}^{168}$ denote the optimized flowrate trajectory generated at hour t . The aligned trajectory is obtained via a circular shift defined as $\tilde{\mathbf{x}}_t = \text{Roll}(\mathbf{x}_t, -((t-1) \bmod 168))$, where $\text{Roll}(\cdot)$ denotes a circular shift operator. This transformation aligns all trajectories to the same weekly reference, enabling direct comparison across optimization steps under the periodic electricity tariff.

Figure 12 illustrates this alignment procedure for the concentrate flowrate. Figure 12(a) shows the aligned 168-hour trajectory generated at the 4th hour of operation. Because this trajectory begins three hours after the fixed weekly reference, the first three entries are wrapped from the end of the vector and highlighted in red vertical line. Figure 12(b) presents the aligned trajectory generated exactly one week later ($t+168$ hours). Owing to the weekly repetition of the tariff structure, the aligned trajectories in Figures 12(a) and 12(b) exhibit strong similarity, indicating consistency of the optimized response across weekly cycles. The same behavior is observed for another representative hour (10th hour of operation), as shown in Figures 12(c) and 12(d).

Figure 13 further examines short-term stability by displaying aligned concentrate flowrate trajectories generated at four consecutive optimization steps (5th, 6th, 7th, and 8th hours of operation). Despite limited training data during these early hours, the aligned trajectories exhibit consistent structure and magnitude after circular alignment. This observation demonstrates that the optimization produces stable and repeatable weekly flowrate schedules even during early-stage operation, supporting the robustness of the digital twin under data-sparse conditions.

Next, the convergence behavior of the flowrate optimization framework was quantitatively assessed using

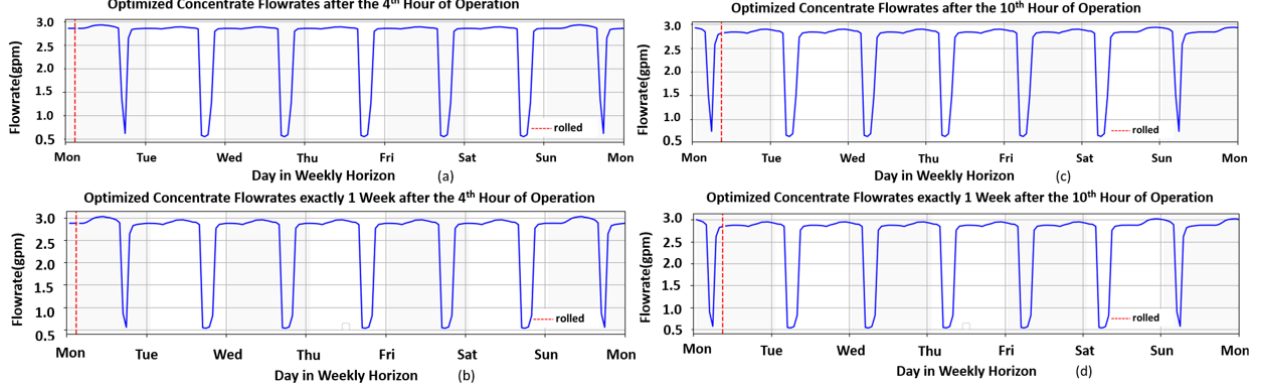


Figure 12: Rolling-aligned optimized concentrate flowrate trajectories (x-axis: Day in Weekly Horizon). (a) Aligned 168-hour trajectory generated at the 4th hour of operation, circularly shifted by three hours relative to the fixed weekly reference (wrapped entries shown in red). (b) Aligned trajectory generated exactly one week after the 4th hour, exhibiting similar structure due to the weekly periodicity of the electricity tariff. (c) Aligned trajectory generated at the 10th hour of operation, circularly shifted by nine hours. (d) Corresponding aligned trajectory generated one week later, showing consistent weekly behavior.

the aligned setpoint trajectories introduced above. Let $\mathbf{x}^{(t)} \in \mathbb{R}^{168}$ denote the aligned 168-hour setpoint trajectory generated at optimization step t for a given flowrate channel, and let $\mathbf{x}^{(t+k)}$ denote the corresponding trajectory generated k hours later.

First, long-term (weekly) convergence was evaluated by comparing trajectories separated by exactly one prediction horizon ($k = 168$). To this end, a normalized root-mean-square (RMS) difference metric was defined as:

$$D_k(t) = \frac{1}{k} \sqrt{\frac{\sum_{i=1}^k \left(x_i^{(t+k)} - x_i^{(t)} \right)^2}{\sum_{i=1}^k \left(x_i^{(t)} \right)^2}},$$

which is dimensionless and quantifies the fractional change in trajectory shape relative to the overall magnitude of the reference trajectory. Values of $D_{168}(t)$ are computed for each optimization step and plotted as a function of time for the concentrate, permeate, and recirculation flowrates in Figures 14(a)–(c). A decreasing trend in $D_{168}(t)$ indicates increasing consistency of the optimized weekly schedules, suggesting convergence of both the model and the associated optimizer under recurring tariff conditions.

To complement this analysis, the short-term (hour-to-hour) convergence was also examined by computing the same normalized RMS metric for trajectories separated by one hour, defined as

$$D_1(t) = \frac{1}{k} \sqrt{\frac{\sum_{i=1}^k \left(x_i^{(t+1)} - x_i^{(t)} \right)^2}{\sum_{i=1}^k \left(x_i^{(t)} \right)^2}}.$$

Figures 15(a)–(c) show the resulting $D_1(t)$ values for the three flowrate channels. As with the weekly comparison, a clear decreasing trend is observed, indicating progressive stabilization of the optimized setpoint trajectories across successive hourly updates.

Taken together, the results in Figures 14 and 15 demonstrate convergence of the optimized flowrate schedules at both weekly and hourly time scales. This multi-scale convergence behavior confirms that the proposed digital twin framework yields stable, repeatable control policies as training data accumulate and the model matures. These results support the feasibility of extending the proposed framework to full scale water reuse facilities, where similar tariff aware operation could yield substantial operational cost savings.

The two-week deployment demonstrates that the proposed digital twin framework achieves stable, interpretable, and economically consistent operation under dynamic electricity tariffs. The optimized trajectories

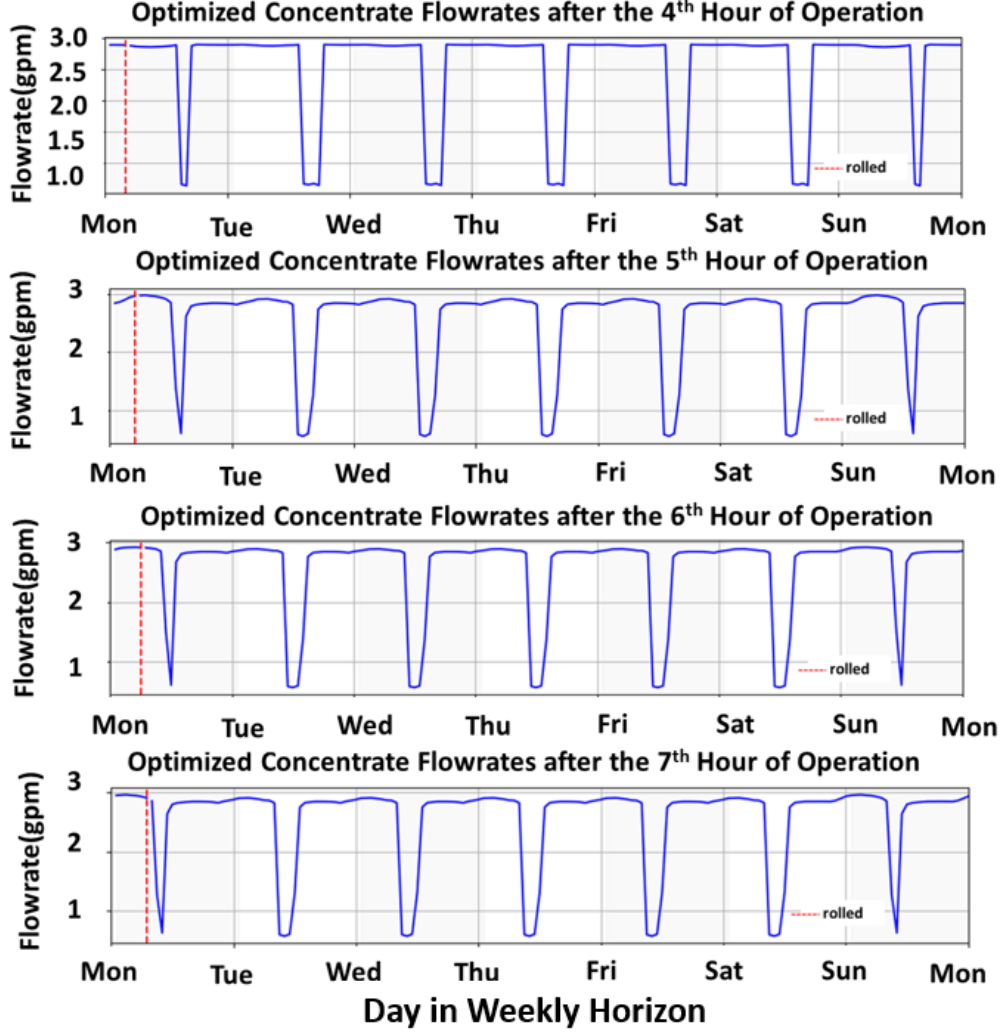


Figure 13: Rolling-aligned optimized concentrate flowrate trajectories for four consecutive optimization steps (5th, 6th, 7th, and 8th hours of operation) (x-axis: Day in Weekly Horizon). Each trajectory is circularly aligned to the fixed weekly reference (Monday 12:00 AM), with wrapped entries indicated by red markers. The strong similarity across consecutive aligned trajectories demonstrates short-term stability of the optimized weekly schedules, even during early-stage operation with limited training data.

exhibit strong weekly periodicity aligned with the tariff structure, indicating that the controller internalizes economic patterns rather than responding myopically to short-term fluctuations. This behavior confirms that the receding horizon successfully translates economic signals into coordinated operational adjustments.

An important open challenge concerns improving computational scalability for long-horizon optimization. The weekly electricity tariff exhibits repeated patterns, with identical tariff values recurring across multiple hours. Under the assumption that the learned model is time-invariant over the prediction horizon, this structural regularity suggests that dimensionality reduction strategies may be possible. Such approaches may become particularly relevant when extending the prediction horizon beyond the current weekly window, when adopting finer temporal resolution for control updates, or when coordinating optimization across multiple treatment units in plant-wide operation. In these settings, the dimensionality of the optimization problem increases substantially. Exploiting the repetitive tariff structure could therefore reduce computational burden and improve scalability. Future research will investigate structured parameterizations of the decision variables that preserve optimality while improving computational efficiency for industrial-scale deployment.

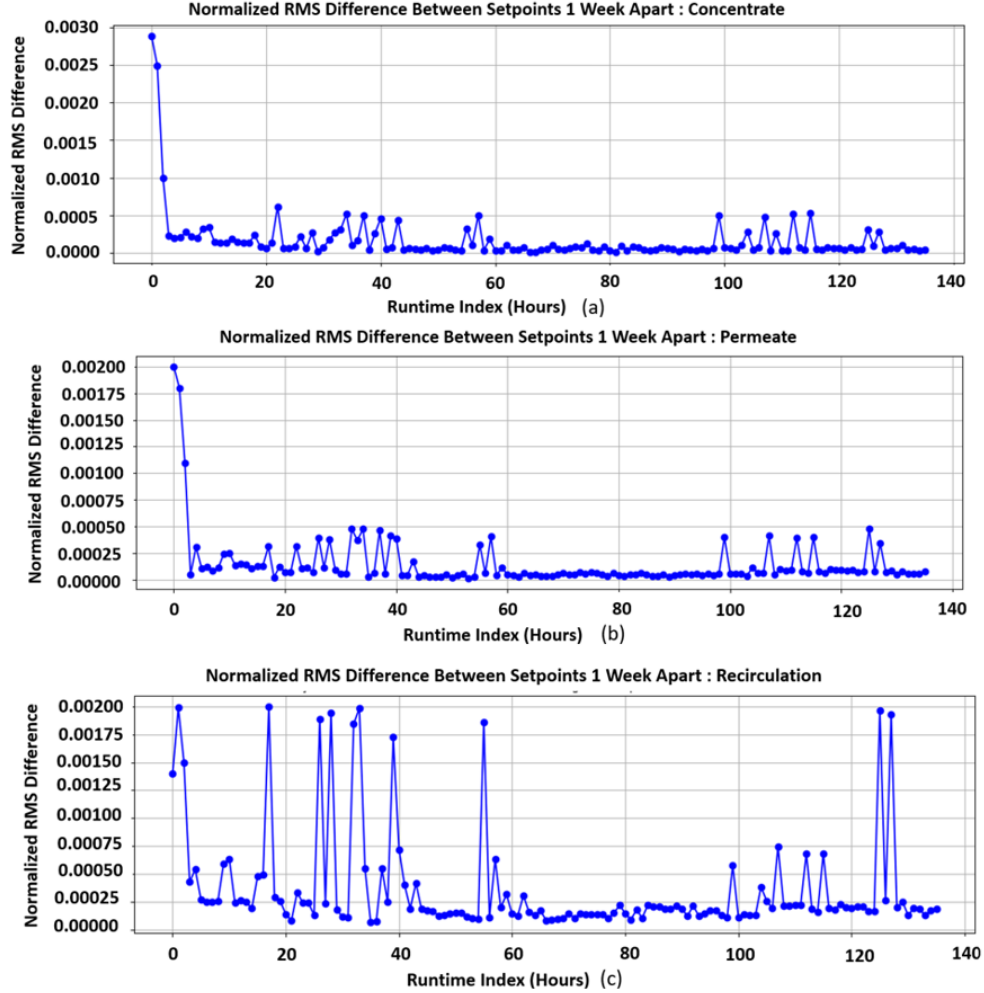


Figure 14: Normalized RMS difference between aligned setpoint trajectories separated by one week ($k = 168$) for (a) concentrate, (b) permeate, and (c) recirculation flowrates.

3.1 Operational Considerations and Utility Perspective

The results presented here demonstrate the potential of digital twins for tariff-responsive optimization in water reuse systems; however, their practical interpretation requires consideration of the operational context in which such tools would be deployed. The framework should not be interpreted as replacing plant operators or removing human oversight from safety-critical treatment operations. Rather, digital twins are more appropriately viewed as decision-support and optimization tools that can operate within a human-in-the-loop framework, where trained operators and utility personnel retain supervisory authority and responsibility for implementation. Depending on the application, the level of automation may range from advisory recommendations to operator-supervised automated execution during selected operating periods (e.g., night-time autonomous).

The magnitude of the observed cost savings is also dependent on the electricity tariff structure assumed in the analysis. The time-of-use scenario considered here was selected to evaluate the potential benefits of tariff-responsive operation, whereas utilities operating under flatter electricity rates may realize smaller economic gains from comparable strategies. Accordingly, the results should be interpreted as demonstrating the potential value of digital twins under tariff structures that provide meaningful economic incentives for time-shifting operation. More broadly, as electricity pricing structures evolve, digital twin frameworks may provide utilities with a quantitative tool to evaluate future operating scenarios under different tariff designs.

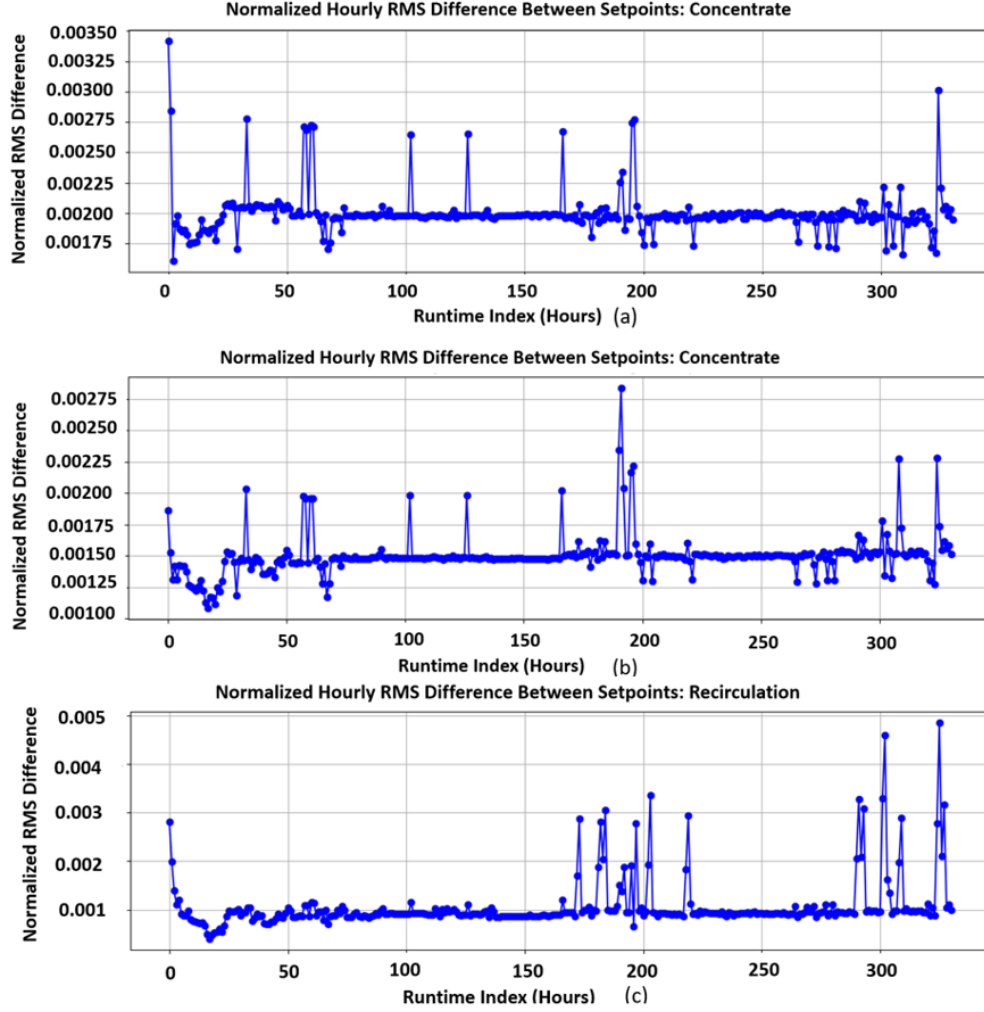


Figure 15: Normalized RMS difference between aligned setpoint trajectories separated by one hour ($k = 1$) for (a) concentrate, (b) permeate, and (c) recirculation flowrates.

Although permeate quality remained within acceptable operating ranges during the study period, the broader operational implications of dynamic flow variation warrant further investigation. In particular, repeated changes in flowrate, recovery, or flux may affect equipment condition, membrane integrity, or long-term operating stability in ways that are not captured within the present two-week validation period. These considerations are especially important for full-scale implementation, where acceptable control strategies must balance economic benefit against mechanical reliability, operational simplicity, and regulatory compliance.

The specific operating strategy explored here may also be more directly applicable to smaller reverse osmosis systems than to large multi-train facilities. In the present pilot, cost-responsive operation was achieved by dynamically adjusting permeate, concentrate, and recirculation flowrates within a single-stage reverse osmosis configuration. In contrast, large full-scale reverse osmosis facilities often consist of multiple parallel trains, for which production changes may be more practically achieved by bringing individual trains online or offline rather than dynamically varying recovery or flux across the system. For example, large facilities with many parallel reverse osmosis units may have sufficient operational flexibility to adjust production through train-level scheduling, whereas smaller systems with only one or two units may have fewer options and may rely more directly on flowrate or recovery adjustments. Thus, while the broader conclusion that digital twins can support economically responsive operation remains valid, the specific control actions demonstrated here

should be interpreted in light of plant scale, configuration, and operational constraints.

4 Conclusion and Future Work

4.1 Conclusion

This work demonstrated a fully automated digital twin framework for online energy cost optimization in a pilot-scale physical representation of an advanced water reuse system that treated real plant water onsite at a full-scale advanced water purification facility. The empirical model-based optimization (EMBO) pipeline integrates continuous model calibration and cost-aware optimization within a closed-loop architecture. Over two weeks of uninterrupted deployment, the framework maintained stable surrogate structure and achieved a 38% reduction in operational (energy) cost relative to nominal steady-state operation, while exhibiting smooth and feasible control updates. Beyond its demonstrated economic performance, the framework provides interpretability through sparse regression, low computational overhead suitable for hourly operation, and resilience to limited or evolving datasets. These properties are essential for scalable implementation in full-scale treatment facilities. This proposed study represents one of the first sustained demonstrations of a digital twin based on empirical modeling operating in closed-loop control with a physical water treatment system.

Broadly, research advancing digital twin applications for the water sector should be designed and implemented in close collaboration with utility staff to ensure relevance to real-world operations, account for the experience and perspectives of plant operators and utility leadership, and identify practical knowledge gaps and research needs. Taken together, the results validate the robustness and reliability of the digital twin in a real pilot environment and demonstrate the potential of such frameworks to support economically responsive operation in water reuse systems.

4.2 Future Work

Future extensions will focus on multi-objective optimization that jointly balances energy consumption, membrane fouling, and water quality constraints. Additional research will explore hybrid digital twins that integrate empirical models with physics-based simulators to enhance interpretability and generalization.

Integration of adaptive tariff forecasting, anomaly detection, and large-scale supervisory control will further advance the framework toward industrial ready, autonomous, human-in-the-loop operation in sustainable water reuse systems. Finally, the long-term effects of dynamic operations on equipment should be investigated.

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AUTHORS' CREDIT STATEMENT

Mukherjee: paper draft, data analysis, coding and curation; Ghorbani Bam: field experiments, coding, paper editing; Melin: data analysis, paper editing; Ghanem: data analysis, paper editing; Rezaei: field experiments, paper editing, research coordination; Gu: field experiments coordination, QA/QC, paper editing; Plumlee: field experiments coordination, QA/QC, paper editing; Tarroja: research supervision, data analysis, paper editing; Rosso: funding acquisition, research coordination, research design, pilot design, paper editing; Villez: research coordination, research design, data analysis, paper editing.

A Appendix

A.1 Physical Pilot Specifications

The physical pilot system was installed at the Orange County Water District (OCWD) Advanced Water Purification Facility and served as the physical counterpart for the digital twin framework. The pilot was designed as a compact, instrumented representation of an advanced water reuse treatment train and included ultrafiltration (UF), an intermediate filtrate/break tank, cartridge filtration, reverse osmosis (RO), and ultraviolet (UV) treatment. Figure A1 shows the installed pilot system and the labeled major components.

The UF subsystem used polymeric hollow-fiber membrane elements (model M-U4040HF09) housed in a PV4040PWG pressure vessel rated at 200 psi. The RO subsystem was configured with four 4 in. $\times$ 40 in. low-energy brackish-water RO membrane elements (model M-T4040ALE) installed in 40E30N-WHITE pressure vessels rated at 300 psi. Unlike the full-scale OCWD facility, which uses a multi-stage RO configuration, the pilot used a single-stage RO configuration with concentrate recirculation to achieve comparable effective recovery within a compact pilot-scale system. Major pilot components are summarized in Table A1.

The pilot included three major pumps: a feed pump upstream of UF, a booster pump upstream of cartridge filtration, and a high-pressure RO pump. The high-pressure RO pump was equipped with variable-frequency-drive control, enabling adjustment of RO operating conditions. The pilot was also equipped with online instrumentation for monitoring flowrate, pressure, pH, electrical conductivity/TDS, and aggregate power demand. These measurements were displayed through the human-machine interface and stored through a cloud-connected monitoring system. The digital twin used online measurements of permeate flowrate, concentrate flowrate, recirculation flowrate, and aggregate pilot power demand for empirical model calibration and energy-cost optimization.

Table A1: Major components of the physical pilot system. Abbreviations and units: UF = ultrafiltration; RO = reverse osmosis; UV = ultraviolet; hp = horsepower; gpm = gallons per minute; psi = pounds per square inch; VFD = variable-frequency drive.

Category	Component	Model / part reference	Key information
Membranes	UF membrane	M-U4040HF09	Quantity: 2
Membranes	RO membrane	M-T4040ALE	Quantity: 4; 4 in. $\times$ 40 in.
Pumps	Feed pump	PND-10	3/4 hp; max flow 12 gpm
Pumps	RO booster pump	Goulds 1ST1D5D4	3/4 hp; max flow 43 gpm
Pumps	RO high-pressure pump	Webtrol XG10B26S16-3TW	3 hp; max flow 16 gpm; VFD controlled
Pressure vessels	UF pressure vessel	PV4040PWG	Rated at 200 psi
Pressure vessels	RO pressure vessels	40E30N-WHITE	Quantity: 4; rated at 300 psi
UV units	UF UV unit	UV-SPH-10U-HP	Capacity: 10 gpm
UV units	RO permeate UV unit	UV-SPH-6	Capacity: 6 gpm

A.2 Convex-Hull-Derived Feasibility Constraints

Table A2 provides the explicit linear inequality coefficients defining the convex polytope used to approximate the empirically realizable operating region described in subsection 2.3. Each inequality corresponds to a supporting half-space of the three-dimensional convex hull of feasible setpoints.

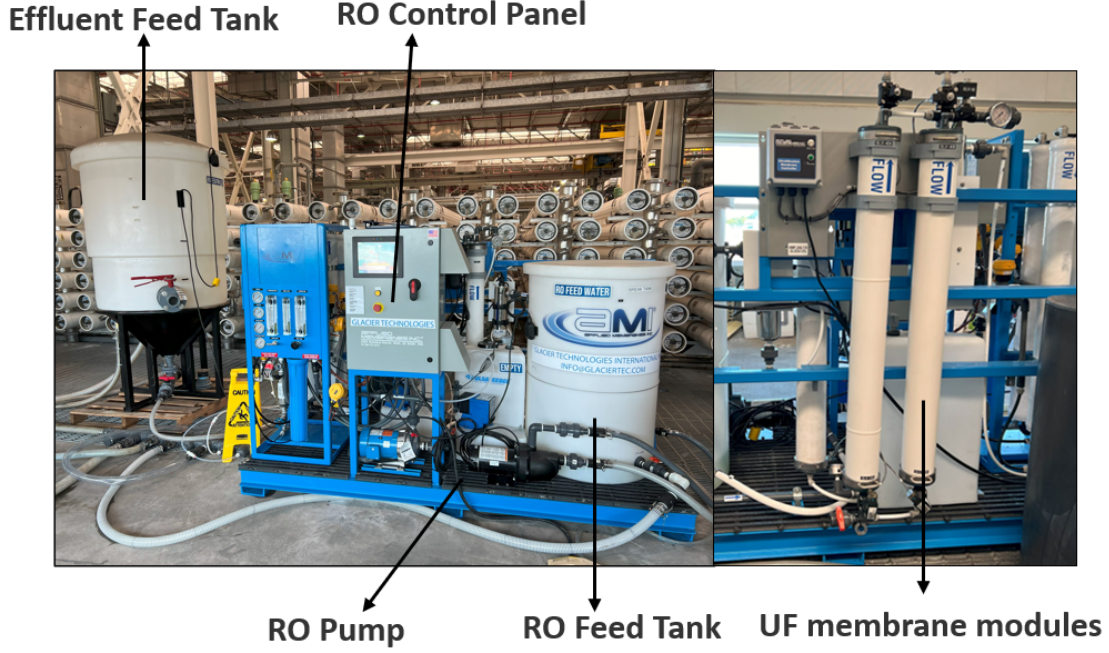


Figure A1: Physical pilot system installed at the Orange County Water District Advanced Water Purification Facility. Labeled components include the effluent feed tank, RO control panel, RO pump, RO feed tank, and UF membrane modules.

Table A2: Linear inequality constraints derived from the 3D convex hull of feasible setpoints. Each row corresponds to a facet: $a_{\text{perm}}F_{\text{perm}} + a_{\text{conc}}F_{\text{conc}} + a_{\text{reyc}}F_{\text{reyc}} \leq b$.

a_{perm}	a_{conc}	a_{reyc}	b	Inequality
0.9570	-0.0166	0.2897	4.1146	$\leq$
1.0000	0.0000	0.0000	3.9000	$\leq$
-0.8830	-0.0859	0.4615	-0.2337	$\leq$
0.8034	0.2373	0.5461	5.1699	$\leq$
-0.6034	0.4872	0.6314	1.8571	$\leq$
-0.7255	0.6829	-0.0854	0.3798	$\leq$
-0.8369	0.5325	-0.1268	-0.7608	$\leq$
-0.8453	0.5265	-0.0908	-0.7183	$\leq$
-0.8346	0.4793	-0.2715	-1.0022	$\leq$
0.9954	-0.0376	-0.0885	3.7054	$\leq$
0.9342	-0.2626	0.2417	3.6139	$\leq$
-0.0097	-0.9959	0.0904	-0.3721	$\leq$
-0.0172	-0.9978	0.0645	-0.4193	$\leq$
0.5358	-0.1478	-0.8313	1.3948	$\leq$
0.2429	-0.3339	-0.9108	0.1518	$\leq$
-0.6059	0.2064	-0.7683	-1.2978	$\leq$
-0.7109	0.3281	-0.6220	-1.2304	$\leq$
-0.1541	-0.1541	-0.9760	-0.7089	$\leq$
-0.5415	0.1692	-0.8235	-1.2116	$\leq$
0.1688	-0.2638	-0.9497	-0.0021	$\leq$
0.2916	-0.1106	-0.9501	0.6173	$\leq$
0.0000	0.0000	-1.0000	0.0000	$\leq$
-0.5774	-0.5774	-0.5774	-1.7321	$\leq$

A.3 Additional Optimization Examples

Figures A2 and A3 provide additional representative optimization outcomes from the 53rd and 300th hours of operation, respectively.

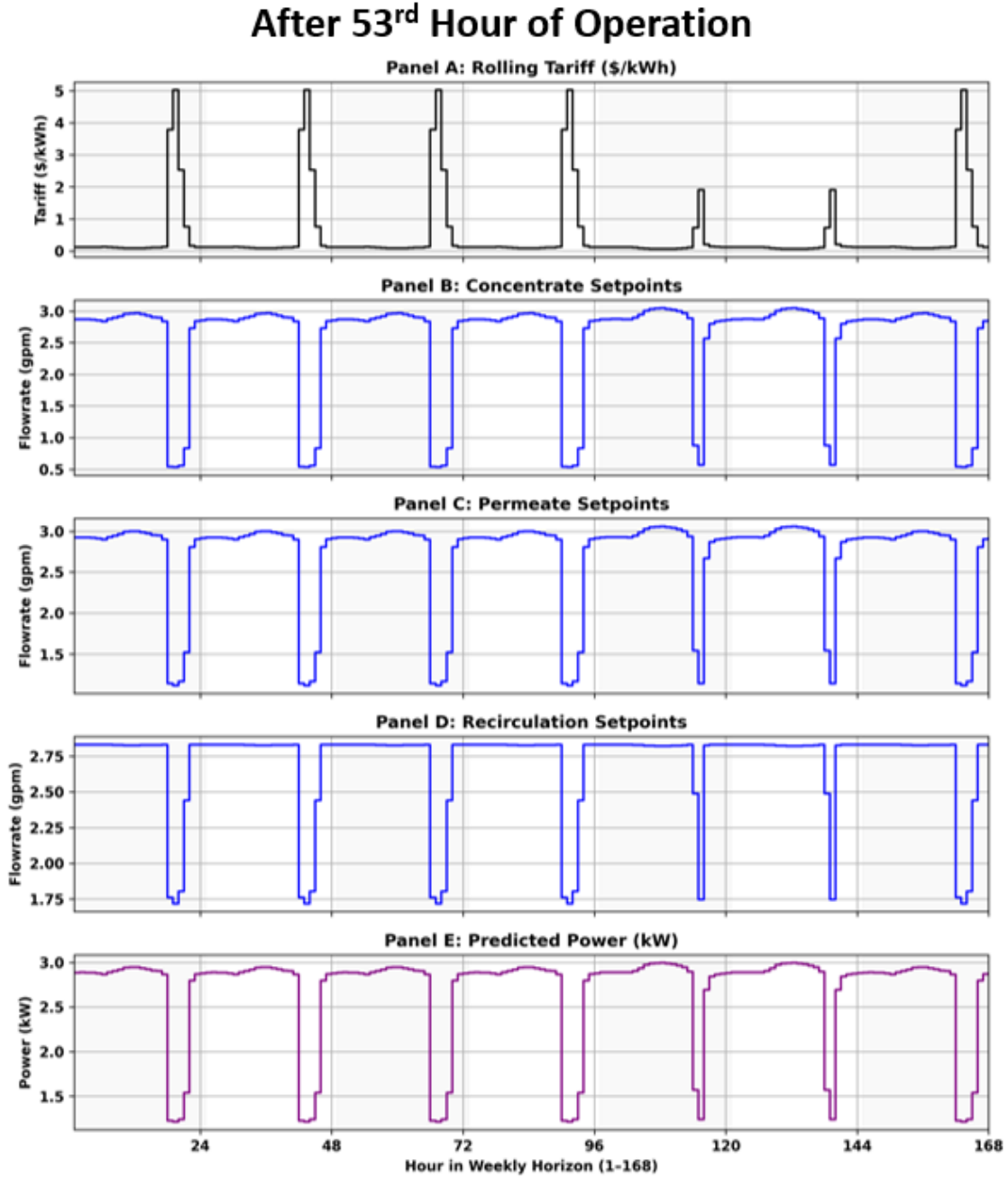


Figure A2: Optimized flowrates, predicted power, and tariff structure at the 53rd hour of operation. The results demonstrate consistent tariff-aware behavior comparable to the representative case discussed in the main text.

After 300th Hour of Operation

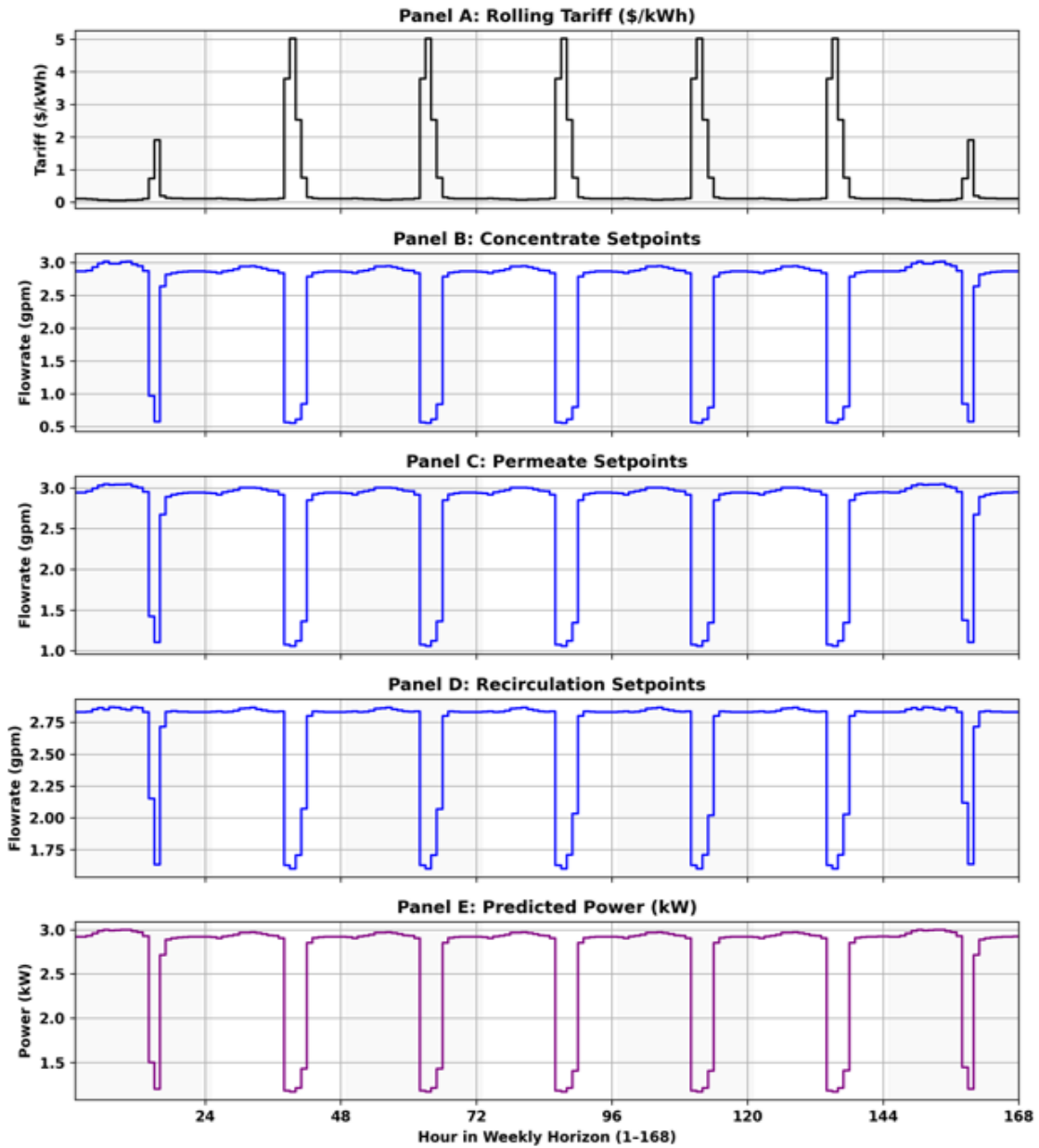


Figure A3: Optimized flowrates, predicted power, and tariff structure at the 300th hour of operation. The optimized schedules exhibit stable and repeatable cost-aware responses under the recurring tariff pattern.

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